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COURSE NAME: MOBILE COMPUTING

Q1 – Wireless Network Architecture and Mobility Analysis (6 Marks)

Part A – Introduction to Wireless Networks

A wireless network connects devices through radio waves instead of requiring a physical cable for every connection. It allows mobile users to communicate while moving within the coverage area. Wireless networks are used in homes, campuses, offices, public places, transportation and emergency services.

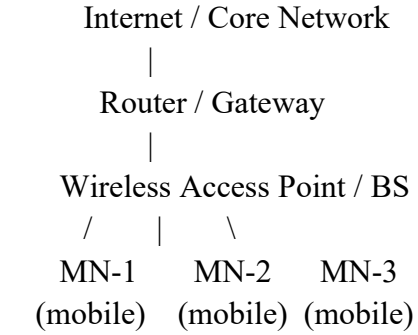
Characteristics of Wireless Networks

- Mobility: users can move while remaining connected, subject to coverage and handoff.
- Shared medium: multiple devices use the same radio channel, so interference and contention may occur.
- Limited bandwidth: wireless capacity is affected by channel width, signal quality, distance and the number of users.
- Variable signal strength: obstacles, distance and interference can reduce signal quality.
- Dynamic connectivity: links may appear or disappear as devices move or conditions change.
- Security requirements: wireless communication needs authentication and encryption to prevent unauthorized access.

Major Components and Their Roles

Component	Role
Mobile node	A smartphone, laptop, tablet, vehicle unit or sensor that sends and receives data.
Access point / base station	Provides radio access and connects wireless devices to the network infrastructure.
Wireless link	Radio communication path between a mobile node and an access point or another node.
Router / gateway	Forwards packets between networks and selects a path toward the destination.
Network server	Provides services such as authentication, applications, storage and address management.
Communication channel	Radio-frequency resource shared by devices for transmitting information.

Wireless Network Architecture



In this infrastructure-based architecture, mobile nodes communicate with the access point or base station. The access point forwards traffic through a router or gateway to other networks. In an ad hoc

network, nodes may communicate directly or forward packets for other nodes without a fixed access point.

Wireless Communication and Constraints

A mobile node first discovers an available access point or neighboring node. It then associates or establishes a link, obtains network configuration where required, and exchanges packets. The access point forwards packets to the destination network. The quality of communication depends on bandwidth, signal-to-noise ratio, interference, congestion, distance and mobility.

- Limited bandwidth can reduce throughput when many users share one channel.
- Weak signal strength can increase retransmissions, delay and packet loss.
- Movement can cause a link to break and require a handoff to another access point.
- Congestion can create queuing delay and reduce the packet delivery rate.

Part B – Wireless Mobility Scenario

Scenario: A student is using a video call while walking from Building A to Building B. Access Point AP-A covers Building A and AP-B covers Building B. As the student moves away from AP-A, the signal becomes weaker. The device discovers AP-B, authenticates or associates with it, and transfers communication to the new point of attachment.

Mobility Event Sequence

1. The mobile node is initially connected to AP-A.
2. The node moves toward the boundary of AP-A coverage.
3. Signal strength from AP-A decreases and packet retransmissions may increase.
4. The node detects AP-B and evaluates its signal quality.
5. The node performs handoff and changes its point of attachment.
6. Communication resumes through AP-B; some packets may be delayed or lost during the transition.

Successful and Interrupted Communication

Condition	Expected behaviour
Successful handoff	AP-B is discovered in time, authentication succeeds and the session continues with a short interruption.
Interrupted handoff	AP-A becomes unavailable before AP-B is ready; packets may be lost and the session may pause.
High mobility	The node crosses coverage boundaries quickly, increasing handoff frequency and signalling.
Poor wireless conditions	Interference or weak signal causes delay, retransmissions and reduced throughput.

Illustrative Mobility Test Results

The following values are sample observations for a worked example, not measurements from a physical experiment. Assume 100 packets are sent in each test.

Test	Mobility condition	Packets received	Packet delivery	Average delay
1	Stationary near AP-A	99/100	99%	20 ms
2	Walking; one handoff	96/100	96%	45 ms
3	Fast movement; repeated handoffs	88/100	88%	85 ms

The sample shows that increasing mobility can reduce packet delivery and increase delay because of handoff signalling, temporary disconnection and route or association updates. Actual values depend on the wireless technology, network load and test environment.

Mobility Decision Logic

```
ALGORITHM ManageWirelessHandoff(MN, CurrentAP, CandidateAP)
INPUT:
   MN – mobile node
   CurrentAP – current access point
   CandidateAP – newly detected access point
OUTPUT:
   Handoff status

BEGIN
   Measure SignalQuality(CurrentAP)
   Discover CandidateAP

   IF CandidateAP does not exist THEN
      RETURN NO_ALTERNATIVE_ACCESS_POINT
   END IF

   Measure SignalQuality(CandidateAP)

   IF SignalQuality(CandidateAP) <= MIN_SIGNAL THEN
      RETURN CANDIDATE_SIGNAL_TOO_WEAK
   END IF

   IF SignalQuality(CandidateAP) > SignalQuality(CurrentAP) THEN
      Authenticate MN with CandidateAP
```

```
      IF Authentication = SUCCESS THEN
         SWITCH attachment TO CandidateAP
         UPDATE network configuration if required
         RESUME data transmission
         RETURN HANDOFF_SUCCESS
      ELSE
         KEEP CurrentAP if it is still available
         RETURN HANDOFF_FAILURE
      END IF
   END IF

   RETURN REMAIN_CONNECTED
END
```

Termination condition: the algorithm ends after a handoff succeeds, the candidate is rejected, or the node remains on the current access point. Repeated movement causes the discovery and decision process to run again.

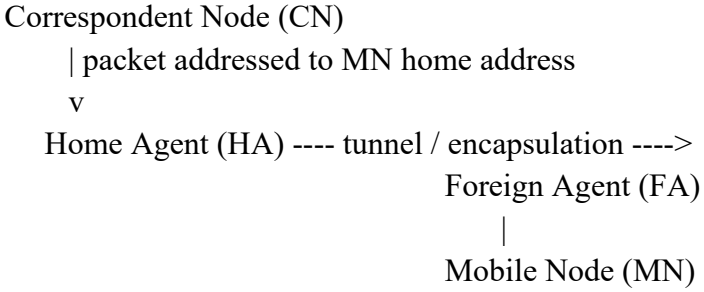
Q2 – Mobile IP and Mobility Management (7 Marks)

Mobile IP Overview

Mobile IP supports communication with a mobile node while it changes its network point of attachment. The mobile node retains a permanent home address. When it visits another network, it obtains a Care-of Address (CoA), which identifies its current location. The Home Agent maintains a binding between the home address and the CoA and forwards packets to the visited network.

Mobile IP Architecture and Entities

Entity	Function
Mobile Node (MN)	Device that moves between networks while retaining its home address.
Home Agent (HA)	Router on the home network that stores the MN-to-CoA binding and tunnels packets.
Foreign Agent (FA)	In traditional Mobile IPv4, assists a visiting MN and may provide a care-of address.
Correspondent Node (CN)	The device or host communicating with the mobile node.
Care-of Address (CoA)	Address representing the mobile node's current point of attachment.



The diagram represents the basic triangular route: packets from the correspondent node reach the home network first, then the Home Agent tunnels them to the current care-of address.

Agent Discovery and Registration

The mobile node discovers mobility agents through advertisements. If it does not receive an advertisement, it may send an agent solicitation. After detecting that it is on a foreign network, it obtains a CoA and sends a registration request to the Home Agent. The request includes the home address, CoA, requested lifetime and an identifier. The Home Agent authenticates the request and returns an acceptance or rejection.

Registration Pseudocode

```
ALGORITHM RegisterMobileNode(MN, HA, CoA)
INPUT:
  MN – mobile node
  HA – home agent
  CoA – current care-of address
OUTPUT:
  Registration result

BEGIN
  Create RegistrationRequest
  Request.HomeAddress ← MN.HomeAddress
  Request.CareOfAddress ← CoA
  Request.Lifetime ← RequestedLifetime
  Request.Identifier ← GenerateRegistrationIdentifier()

  Authenticate Request
  SEND Request TO HA
  WAIT FOR RegistrationReply

  IF RegistrationReply.Status = ACCEPTED THEN
    Binding[MN.HomeAddress] ← CoA
    RegistrationExpiry ← CurrentTime + RegistrationReply.Lifetime
    RETURN REGISTRATION_SUCCESS
  ELSE
    RETURN REGISTRATION_FAILURE
  END IF
END
```

Packet Delivery and Tunnelling

7. The Correspondent Node sends a packet to the mobile node's permanent home address.
8. Normal routing delivers the packet to the home network.
9. The Home Agent checks its binding table and finds the current CoA.
10. The Home Agent encapsulates the original packet inside a new packet addressed to the CoA.

11. The tunnel carries the encapsulated packet to the visited network or Foreign Agent.
12. The tunnel endpoint decapsulates the packet and delivers the original packet to the Mobile Node.

Packet Tunnelling Pseudocode

```
ALGORITHM TunnelPacketToMobileNode(Packet, DestinationHomeAddress)
INPUT:
  Packet – original IP packet
  DestinationHomeAddress – mobile node home address
OUTPUT:
  Delivery status

BEGIN
  IF Binding[DestinationHomeAddress] does not exist THEN
    DROP Packet
    RETURN DELIVERY_FAILURE
  END IF

  CoA ← Binding[DestinationHomeAddress]
  EncapsulatedPacket ← ENCAPSULATE(Packet, CoA)
  FORWARD EncapsulatedPacket TO CoA

  IF TunnelDeliverySuccessful THEN
    Decapsulate at tunnel endpoint
    DELIVER original Packet TO MobileNode
    RETURN DELIVERY_SUCCESS
  ELSE
    RETURN DELIVERY_FAILURE
  END IF
END
```

Mobile IP Challenges

Challenge	Explanation	Possible reduction
Registration delay	The mobile node must discover an agent, obtain a CoA and register it.	Use efficient discovery, reduce unnecessary registrations and prepare the target network early where possible.
Tunnelling overhead	Encapsulation adds an outer header and consumes extra bandwidth.	Use efficient tunnelling and avoid unnecessary encapsulation where a safe route optimisation is supported.
Triangular routing	Packets may travel CN → HA → MN even when a shorter direct path exists.	Use route optimisation when supported and secure.
Packet loss during handoff	Packets may arrive at the old location or be lost while the	Use buffering, retransmission, forwarding from the old agent

	binding is updated.	or make-before-break handoff where supported.
Security	False registration could redirect traffic to an attacker.	Authenticate registration messages and protect mobility signalling.
Scalability	A large number of mobile nodes creates binding and signalling load at agents.	Distribute load, use efficient state management and limit unnecessary updates.

Joint Design: Frequent Network Changes During an Active Session

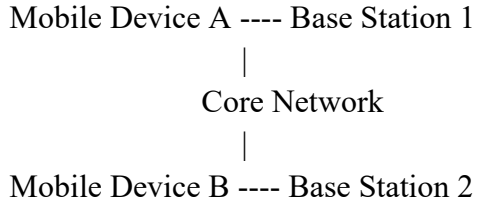
Frequent movement can cause repeated agent discovery, CoA acquisition and registration. During each handoff, packets may be delayed, duplicated or lost. Tunnelling adds header overhead, while triangular routing can increase end-to-end delay. If the mobile node changes networks faster than registration completes, its binding may become outdated and packets may be sent to the wrong location.

These problems can be reduced by selecting a stable target network using signal quality and hysteresis, preparing the new connection before releasing the old one, buffering packets during handoff, using suitable registration lifetimes, authenticating updates, and applying route optimisation where available. The network should also monitor delay, packet loss and registration failures. These measures reduce interruption but cannot guarantee zero delay in every wireless environment.

Q3 – Cellular Networks versus Ad Hoc Wireless Networks (7 Marks)

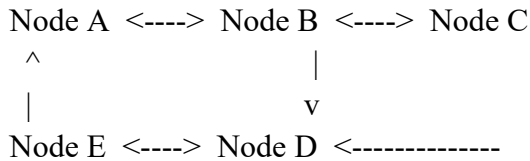
Cellular Network Architecture

A cellular network divides a geographical area into cells. Each cell is served by a base station. Mobile devices communicate with the base station, which connects them to the operator's core network. The network manages authentication, resource allocation, mobility and handoff between cells.



Ad Hoc Network Architecture

An ad hoc wireless network is formed by wireless nodes without depending on fixed base stations or access points. Nodes communicate directly when they are within range. If the destination is farther away, intermediate nodes forward packets using multi-hop routing. The topology can change whenever nodes move or links fail.



Technical Comparison

Parameter	Cellular Network	Ad Hoc Network
Infrastructure	Uses base stations and operator/core infrastructure.	No fixed infrastructure is required.
Communication	Mobile device normally communicates through a base station.	Nodes communicate directly or through intermediate nodes.
Routing	Infrastructure-assisted and centrally managed in the network.	Distributed multi-hop routing among nodes.
Mobility	Network performs handoff between cells.	Routes and neighbor relationships change as nodes move.
Topology	Planned cell-based coverage.	Dynamic topology.
Scalability	Can support large user populations with planned capacity and infrastructure.	Scalability depends on node density, radio range, routing overhead and mobility.
Handoff	Managed by cellular network procedures.	Connectivity changes through neighbor and route discovery/maintenance.
Failure handling	Base-station or core failure can affect a coverage area or service.	Alternate peer routes may be available if nodes remain connected.
Applications	Public mobile voice/data, broadband and fleet connectivity.	Disaster response, military, temporary events and field networks.

Joint Engineering Decision: Disaster Environment

For an emergency or disaster area where fixed infrastructure is unavailable, an ad hoc wireless network is the more suitable primary architecture. Rescue-team radios, vehicle units and portable devices can form a temporary multi-hop network without waiting for base stations to be restored. Each node can forward packets for other nodes, extending communication beyond the direct radio range.

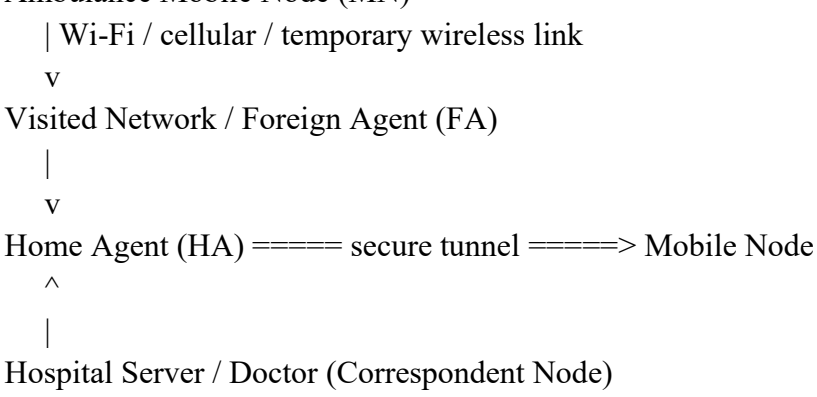
The design should use multiple paths where possible, prioritize emergency messages, monitor link quality and provide gateways to any surviving cellular or satellite connection. Ad hoc networks also have limitations: bandwidth is shared, battery power is limited, routes may break frequently and security must be managed. Therefore, a hybrid design that uses ad hoc communication locally and surviving infrastructure as an optional gateway is preferable when available.

Q4 – Applications of Mobile IP and Routing Requirements (7 Marks)

Selected Application: Emergency Ambulance and Healthcare Communication

Mobile IP can support an ambulance that moves between hospitals, emergency sites and different wireless networks while maintaining access to medical services. The ambulance may send patient vital signs, video, location, diagnostic information and transport status to a hospital control centre. The ambulance is the Mobile Node, the hospital or service network contains the Home Agent, the visited network provides connectivity and a hospital server or doctor acts as the Correspondent Node.

Application Network Architecture



Communication Flow

13. The ambulance connects to an available wireless network and discovers the mobility agent.
14. It obtains a CoA for the visited network.
15. The ambulance registers the CoA with its Home Agent.
16. The hospital server sends data to the ambulance's permanent home address.
17. The Home Agent tunnels packets to the current CoA.
18. The ambulance decapsulates and receives the data; it can also send patient updates to the hospital.
19. When the ambulance moves again, it obtains a new CoA and updates its registration.

Mobile Entities and Requirements

Entity / requirement	Description
Ambulance unit	Mobile node carrying patient monitoring and communication equipment.
Hospital control centre	Correspondent endpoint receiving patient data and sending instructions.
Home Agent	Maintains the mobile node's location binding and forwards packets.
Visited network	Provides the current wireless point of attachment and CoA.
Low delay	Critical alerts and doctor instructions should arrive quickly.
Reliable delivery	Patient data should be acknowledged, buffered or retransmitted when necessary.
Security and privacy	Authenticate devices and protect patient information using encryption and access control.

Availability	Provide alternate wireless links or store-and-forward operation during temporary outages.

Performance and Suitability Analysis

Mobile IP is useful because the ambulance can retain a stable home address while changing its point of attachment. However, registration and tunnelling can add delay and overhead. A handoff may interrupt a live video stream or delay a vital-sign update. Critical alerts should therefore receive priority, and the application should use buffering and retransmission for important data. Real-time video can adapt its bit rate when bandwidth falls.

Mobile IP is suitable when session continuity and a stable address are important. It does not itself guarantee low latency, bandwidth or packet delivery. The wireless access network, routing protocol, security mechanisms and application design must work together to meet emergency requirements.

Routing Requirements

- Routes must be discovered or maintained as vehicles move.
- The network should select paths with acceptable delay, packet loss and hop count.
- Link failures must be detected quickly and repaired or rerouted.
- Emergency traffic should be prioritized over non-critical traffic.
- Control overhead should be limited because wireless bandwidth and battery power are constrained.

Q5 – Proactive Routing Protocols in Mobile Networks (6 Marks)

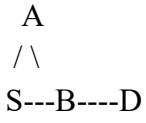
Proactive Routing Concept

A proactive routing protocol maintains routes to destinations before data transmission is requested. Nodes periodically exchange routing information and update their routing tables. When an application sends a packet, a route may already be available, reducing initial route-discovery delay. The disadvantage is periodic control traffic, even when some routes are not being used.

DSDV Operation

Destination-Sequenced Distance Vector (DSDV) is a table-driven proactive routing protocol. Each node maintains routing information containing destination, next hop, hop count and destination sequence number. Sequence numbers indicate the freshness of route information and help prevent routing loops. A route with a newer sequence number is preferred. If sequence numbers are equal, a route with a smaller hop count may be selected.

Sample Topology



Assume S needs to reach D. Initially, S may know a route S → B → D with two hops. If the S–B link fails but S can reach A and A can reach D, updated information may provide S → A → D. Nodes exchange updates and replace stale route information according to sequence numbers and route metrics.

DSDV-Style Route Update Pseudocode

```
ALGORITHM UpdateDSDVRoute(Node, ReceivedRoute)
INPUT:
  Node – receiving node
  ReceivedRoute = {Destination, NextHop, HopCount, SequenceNumber}
OUTPUT:
  Updated routing table

BEGIN
  Destination ← ReceivedRoute.Destination
  NewHopCount ← ReceivedRoute.HopCount + 1
  NewSequence ← ReceivedRoute.SequenceNumber

  IF Destination not in RouteTable THEN
    ADD route with NextHop = Node.Neighbor
    SET HopCount = NewHopCount
    SET SequenceNumber = NewSequence
  ELSE IF NewSequence > RouteTable[Destination].SequenceNumber THEN
    REPLACE RouteTable[Destination]
  ELSE IF NewSequence = RouteTable[Destination].SequenceNumber AND
       NewHopCount < RouteTable[Destination].HopCount THEN
    REPLACE RouteTable[Destination]
  END IF

  RETURN RouteTable
END
```

Routing Table Example

Node	Destination	Next hop	Hop count	Sequence number
S	D	B	2	18
A	D	D	1	18
B	D	D	1	18
D	D	D	0	18

After the S–B link fails, S can use an updated route through A if A advertises a valid, sufficiently fresh route. The precise table update depends on the sequence number and metric received.

Performance Investigation

Parameter	Effect in proactive routing
Route availability	Routes are generally available before data transmission.
Initial delay	Usually low because route discovery is not needed for each new flow.
Control overhead	Periodic updates consume bandwidth and energy.
Convergence	After topology changes, nodes need time to

	receive and process updates.
Network size	Larger networks can create more routing information and update traffic.
Mobility	Frequent changes can cause frequent updates and temporary stale routes.

DSDV is appropriate when traffic is frequent and low route setup delay is important, and when topology changes are not so rapid that periodic updates overwhelm the network. It is less efficient for sparse, intermittent communication where many maintained routes are never used.

Q6 – Reactive Routing and Protocol Comparison (7 Marks)

Reactive Routing Concept

A reactive routing protocol discovers a route only when a source needs to communicate and no valid route is available. This demand-driven approach reduces periodic routing traffic. However, the first packet may wait while route discovery takes place.

AODV Operation

Ad hoc On-Demand Distance Vector (AODV) is a reactive routing protocol. The source broadcasts a Route Request (RREQ). Intermediate nodes record a reverse route to the source and forward the request if they have not already processed it. The destination, or an intermediate node with a valid route, sends a Route Reply (RREP) back along the reverse path. When a link failure affects an active route, a Route Error (RERR) informs affected nodes and the source may start a new route discovery.

AODV Message Types

Message	Purpose
RREQ	Discovers a route from source to destination.
RREP	Provides route information and establishes the forward path.
RERR	Reports an unreachable destination or broken route.

AODV Route Discovery Pseudocode

```
ALGORITHM ReactiveRouteDiscovery(Source, Destination)
INPUT:
   Source – requesting node
   Destination – required destination
OUTPUT:
   Route or ROUTE_NOT_FOUND

BEGIN
   IF RouteTable[Destination].State = VALID THEN
      RETURN RouteTable[Destination]
   END IF

   RouteRequestID ← RouteRequestID + 1
```

```
    Create RouteRequest(Source, Destination, RouteRequestID)
    Broadcast RouteRequest to neighbors
    WAIT for RouteReply or RouteFailure

    IF RouteReply received THEN
      Establish reverse path
      Establish forward path
      Mark RouteTable[Destination] as VALID
      RETURN RouteTable[Destination]
    ELSE
      RETURN ROUTE_NOT_FOUND
    END IF
  END
```

Route Request Processing

```
  ALGORITHM ProcessRouteRequest(Node, Request)
  INPUT:
    Node – current receiving node
    Request = {Source, Destination, RequestID, HopCount}
  OUTPUT:
    Route discovery action

  BEGIN
    IF Request already processed THEN
      DISCARD Request
      RETURN
    END IF

    Record reverse route toward Request.Source
    Mark Request as processed

    IF Node = Request.Destination THEN
      Create RouteReply
      SEND RouteReply toward Source
      RETURN
    END IF

    IF ValidRouteTo(Request.Destination) THEN
      Create RouteReply using valid route
      SEND RouteReply toward Source
    ELSE
      Request.HopCount ← Request.HopCount + 1
      FOR each Neighbor in NeighborTable DO
```

```
            FORWARD Request TO Neighbor
        END FOR
    END IF
END
```

Route Maintenance and RERR

```
ALGORITHM MaintainReactiveRoute(Source, Destination)
INPUT:
    Source and Destination
OUTPUT:
    Route status

BEGIN
    MONITOR active route

    IF LinkFailureDetected THEN
        Mark failed link as INVALID

        IF AlternateValidRoute exists THEN
            Select alternate route
            RETURN ALTERNATE_ROUTE_USED
        ELSE
            Generate RouteError
            SEND RouteError toward affected source

            IF Source still requires communication THEN
                Initiate new RouteDiscovery
            ELSE
                RETURN ROUTE_TERMINATED
            END IF
        END IF
    END IF

    RETURN ROUTE_ACTIVE
END
```

AODV versus DSDV

Feature	AODV (Reactive)	DSDV (Proactive)
Route discovery	On demand using RREQ/RREP.	Routes maintained continuously through updates.
Routing table	Stores active and discovered route information.	Maintains destination routes in routing tables.
Control overhead	Usually lower when traffic is infrequent; discovery causes bursts.	Periodic updates create continuous overhead.

Initial communication delay	May be higher because route discovery is required.	Usually lower when a valid route is already present.
Mobility handling	Broken active routes trigger repair or rediscovery.	Topology changes are propagated through table updates.
Suitable scenario	Intermittent traffic and dynamic networks where not all destinations are used.	Frequent traffic and conditions where low route setup delay is important.

Joint Question: Highly Dynamic Network

For a highly dynamic mobile network with frequent topology changes and intermittent traffic, AODV is generally the better initial choice. It discovers routes when required and avoids continuously advertising every destination route. This can reduce background control overhead when communication is not constant. When a route breaks, AODV detects the failure and uses an alternate route if available or initiates a new discovery.

The trade-off is route discovery delay. If the application sends continuous high-rate traffic and requires very low initial delay, DSDV may be preferable because routes are maintained in advance. The final selection must consider traffic rate, mobility rate, available bandwidth, acceptable delay and node energy.

Q7 – Design a Mobile Emergency Communication Network (10 Marks)

Scenario and Design Objectives

During a natural disaster, ambulances, police vehicles, rescue teams and medical personnel move across different areas. Fixed communication infrastructure may be damaged or unavailable. The proposed network must provide communication with minimum practical delay, support mobility, tolerate link and node failures, and deliver critical messages reliably.

Stage 1 – Independent Design Proposals

Student A – Network Architecture and Mobility Proposal

- Use a wireless ad hoc mesh as the local communication layer because it can operate without fixed base stations.
- Equip each ambulance, police vehicle and rescue team with a mobile node that can send and forward packets.
- Use Mobile IP when a mobile unit must retain a stable home address while moving between available IP networks.
- Use a gateway connected to a surviving cellular, satellite or emergency operations network when available.
- Prioritize patient alerts, rescue requests and command messages over routine status updates.

Student B – Routing and Failure-Handling Proposal

- Use AODV as the primary routing protocol for the dynamic ad hoc network because routes are discovered on demand.

- Use RREQ and RREP to establish paths and RERR to report broken links.
- Monitor active links and maintain alternate routes where possible.
- Use store-and-forward buffering when a node becomes temporarily disconnected.
- Evaluate packet delivery ratio, end-to-end delay, routing overhead and route recovery time.

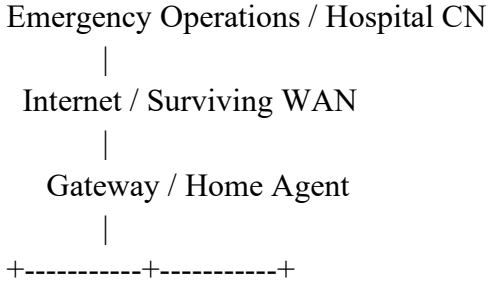
Stage 2 – Comparison of the Two Designs

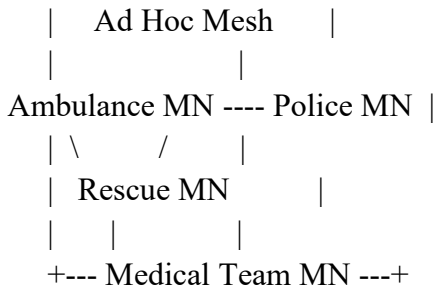
Aspect	Student A proposal	Student B proposal	Joint decision
Main focus	Architecture and mobility continuity.	Route discovery and repair.	Combine an ad hoc mesh with Mobile IP where stable IP mobility is required.
Network structure	Mobile nodes, optional gateways and mobility agents.	Distributed multi-hop forwarding.	Each mobile node participates in the mesh; gateways connect to external networks.
Mobility	Mobile IP registration and handoff.	AODV route repair after topology changes.	Use Mobile IP for inter-network movement and AODV for local ad hoc paths.
Overhead	Registration and tunnelling overhead.	RREQ/RREP/RERR control overhead.	Limit unnecessary registration and route discovery; prioritize critical data.
Delay	Handoff and registration may add delay.	Initial route discovery may add delay.	Use buffering, route caching and pre-established alternate links where supported.
Failure handling	Switch to another available network or gateway.	Invalidate broken route and discover another path.	Use alternate mesh paths, gateways and store-and-forward.

Stage 3 – Joint Final Network Design

The final design uses a hybrid emergency network. Within the disaster area, mobile nodes form an ad hoc mesh. Each node can act as both an end device and a router. AODV discovers paths between nodes when communication is needed. One or more gateway nodes connect the mesh to surviving cellular, satellite or command-centre infrastructure. Mobile IP is used for devices that need to preserve a stable IP identity while changing between IP networks.

Final Architecture Diagram





The mesh links are wireless and can change as vehicles move. The gateway is optional if external infrastructure is unavailable; local communication can continue within the connected mesh.

Mobile IP Registration and Packet Flow

20. A mobile unit detects a new IP network or mobility agent.
21. It obtains a care-of address for the current network.
22. It sends a registration request to its Home Agent and receives an acceptance.
23. The Correspondent Node sends packets to the unit's home address.
24. The Home Agent tunnels packets to the registered CoA.
25. The unit receives the packet and continues communication; on moving again, it updates the binding.

Selected Routing Protocol and Justification

AODV is selected for local ad hoc routing. Emergency units may move frequently, making the network topology dynamic. AODV creates routes only when communication is needed, which can reduce periodic control traffic compared with DSDV when traffic is intermittent. RREQ discovers a route, RREP confirms it, and RERR reports a broken route. The cost is route discovery delay, so critical applications should use suitable route lifetimes, buffering and alternate paths where available.

Failure Handling

Failure	Response
Wireless link failure	Mark the link invalid; use an alternate route if available; otherwise initiate AODV route discovery.
Mobile node failure	Stop forwarding through the failed node and select another path.
Gateway failure	Use another gateway if available; otherwise continue local mesh communication and buffer external messages.
Temporary network disconnection	Store critical messages and forward them when connectivity returns.
Home Agent unreachable	Continue local ad hoc communication; queue external traffic and retry registration when the Home Agent becomes reachable.
Congestion	Prioritize emergency packets, limit non-critical traffic and select a less congested route if possible.

Stage 4 – Validation Plan

The following is a proposed experimental plan. The results must be measured in a simulator or test network before they are reported as actual experimental findings.

Test case	Nodes	Mobility	Traffic	Purpose
T1	5	Low	Periodic status messages	Establish baseline delay and delivery.
T2	10	Medium	Mixed emergency and status traffic	Observe impact of additional nodes.
T3	20	High	Intermittent messages	Evaluate route discovery and frequent topology changes.
T4	20	High	Continuous traffic	Measure control overhead and packet loss under load.
T5	10	Medium	Force one link and one gateway failure	Check route repair and gateway failover.

Performance Metrics and Formulas

Packet Delivery Ratio (PDR) = (Number of packets received / Number of packets sent) × 100.

Average End-to-End Delay = Sum of packet delivery delays / Number of packets received.

Routing Overhead = Number of routing control packets transmitted. It may also be reported as control-packet transmissions per data packet delivered.

Route Recovery Time = Time from detecting a route failure until a valid replacement route is available.

Illustrative Evaluation Table

The values below are hypothetical examples showing how results can be presented. Replace them with simulator output if the assignment requires practical experimental evidence.

Scenario	Packets sent	Packets received	PDR	Average delay	Control packets
5 nodes, low mobility	500	485	97.0%	35 ms	90
10 nodes, medium mobility	500	465	93.0%	58 ms	145
20 nodes, high mobility	500	430	86.0%	95 ms	260

Interpretation: As the number of nodes and mobility increase, the sample results show lower packet delivery, higher delay and more routing control messages. Frequent link changes trigger route repair

and discovery. The actual trend and magnitude depend on node speed, radio range, traffic load, topology and simulator configuration.

Stage 5 – Cross-Defence

Why is AODV appropriate? It discovers routes only when required, which is useful when emergency communication is intermittent and topology changes frequently. It can react to link failures through RERR and route rediscovery. Its main limitation is the delay caused by discovering a route after a break.

How does Mobile IP maintain communication during movement? The mobile node retains its home address and registers a new CoA with its Home Agent when it moves to a visited network. The Home Agent updates the binding and tunnels packets to the current CoA, allowing correspondents to continue addressing the mobile node using its home address.

Stage 6 – Joint Evaluation

- The ad hoc mesh is suitable because it can be deployed without fixed base stations and supports multi-hop communication.
- Mobile IP is useful for maintaining a stable IP identity when a unit changes IP networks; it is not necessary for every local mesh exchange.
- AODV is suitable for on-demand communication in a dynamic topology, although route discovery can add delay.
- Overhead comes from Mobile IP registration and tunnelling as well as AODV route-control messages.
- Mobility can break links, trigger handoffs and increase packet loss and delay.
- When connectivity is lost, local communication may continue in separate network partitions; critical external messages should be buffered and forwarded after reconnection.
- The solution may be unsuitable when nodes are extremely sparse, battery power is critically low, radio interference is severe, or strict real-time delivery cannot tolerate route discovery and disconnection.

Final Design Conclusion

Before selecting Mobile IP and a routing protocol for a highly mobile wireless environment, a designer should analyse infrastructure availability, node speed, topology-change frequency, traffic rate, delay tolerance, bandwidth, radio range, packet loss, energy limits, security, scalability and failure recovery. Mobile IP addresses location changes across IP networks, while routing protocols determine paths within the current network. Their roles are complementary, and the design must evaluate both mobility-management overhead and routing performance.